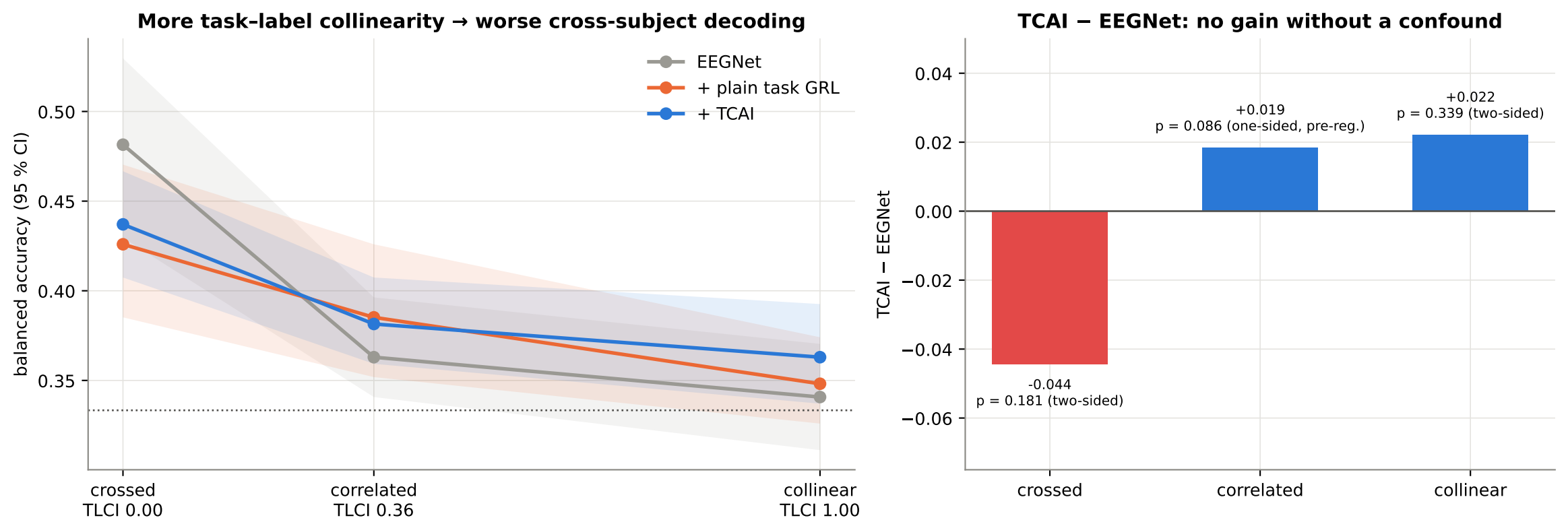
More task–label collinearity → worse cross-subject decoding
EEGNet
+ plain task GRL
+ TCAI
balanced accuracy (95 % CI)
0.50
0.45
0.40
0.35
crossed
TLCI 0.00
correlated
TLCI 0.36
collinear
TLCI 1.00
TCAI − EEGNet: no gain without a confound
TCAI − EEGNet
0.04
0.02
0.00
−0.02
−0.04
−0.06
-0.044
p = 0.181 (two-sided)
+0.019
p = 0.086 (one-sided, pre-reg.)
+0.022
p = 0.339 (two-sided)
crossed
correlated
collinear